

Name:

Netid:

1.) How does IDS simulate BFS behaviour with a DFS?

2.) What is the difference between uninformed and informed search algorithms? Give 3 examples of uninformed search algorithms and 1 example of an informed search algorithm.

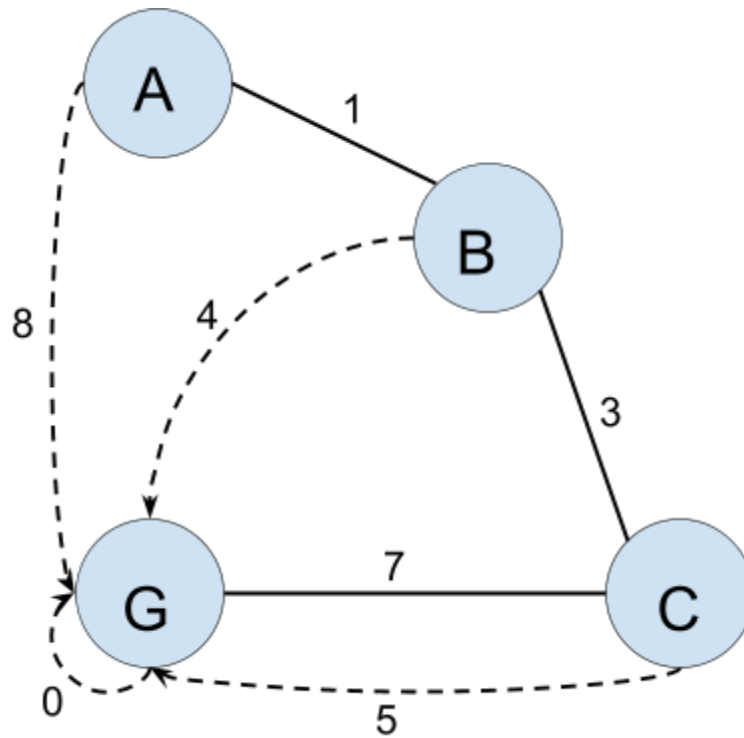
3.) Explain some limitations of BFS and DFS (Tom's Thoughts: What are other search algorithms that can be used to improve upon the limitations of BFS and DFS? Why are these algorithms improvements?) (Tom's Thoughts: How does IDS simulate BFS behaviour with a DFS?)

4.) How could search be construed as emulating human thought?

5.) What is the difference between Dijkstra's and A*?

6.) What is a heuristic function (general definition)? What is a heuristic function in the context of A* Search? What 2 characteristics do you want your heuristic function in A* search to satisfy?

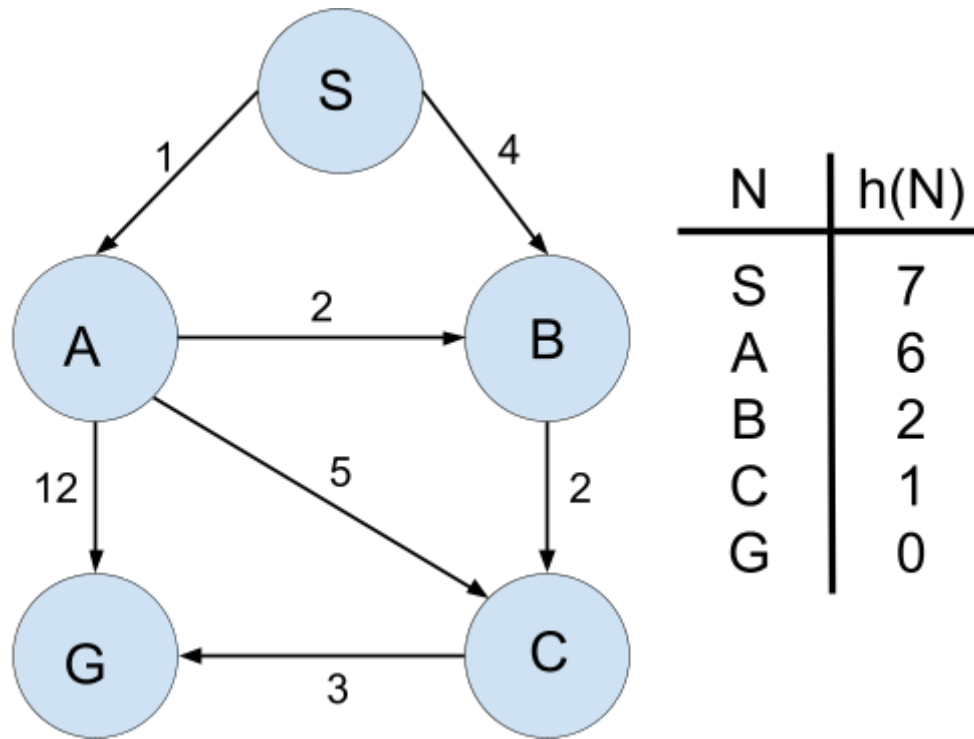
7.) For the graph given below answer the questions. The cost to travel between nodes are given as/on the solid lines and the heuristic distances are on the dashed lines.



a.) Is this heuristic admissible? Why or why not?

b.) Is this heuristic consistent? Why or why not?

8.) Given the following graph, starting in Node S and trying to get to Node G, trace the steps the A* Algorithm would take. To make the graph less busy, the heuristic distances are given as a table.



*.) $N \rightarrow N' = d(N, N') + h(N', G)$

1.) $S \rightarrow A = 1 + 6 = 7$

2.) $S \rightarrow B = 4 + 2 = 6$

3.)

4.)

5.)

6.)

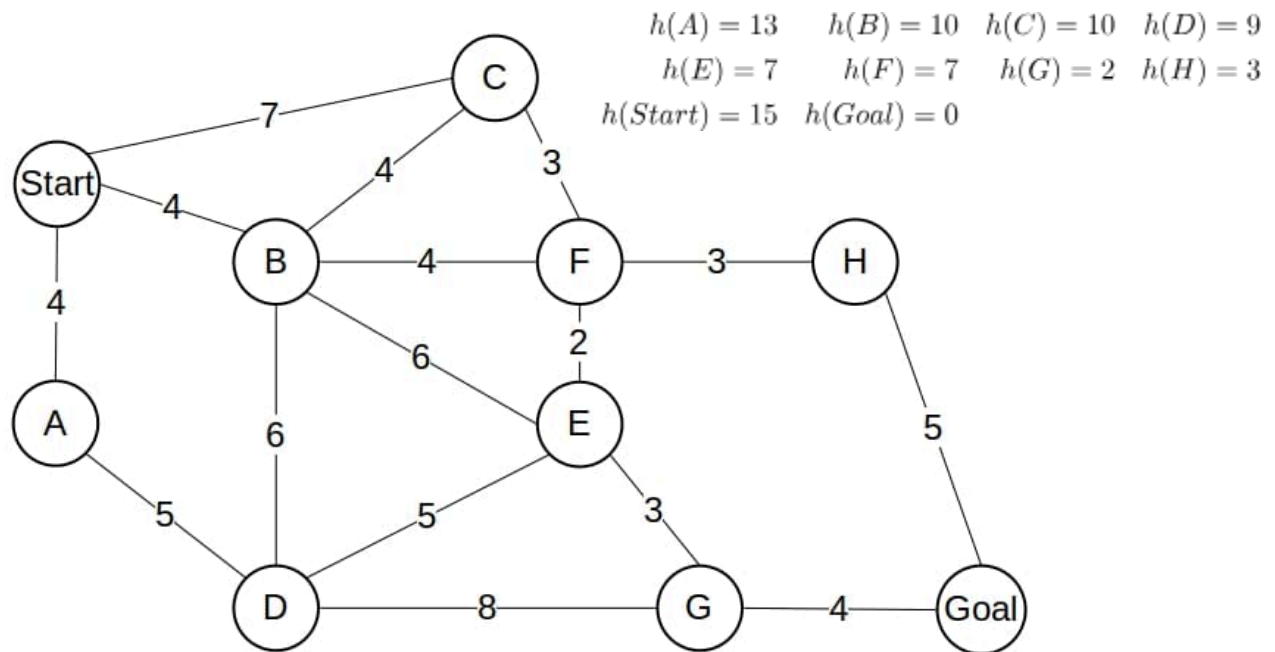
7.)

8.)

9.)

10.)

9.) Given the following graph, starting in Node S and trying to get to Node G, trace the steps the A* algorithm would take. To make the graph less busy, the heuristic distances are given as a table. Much like on Exam 01, my solution uses all 11 spots.



*) $N \rightarrow N' = d(N, N') + h(N', G)$

1.) $S \rightarrow A = 4 + 13 = 17$

2.) $S \rightarrow B = 4 + 10 = 14$

3.) $S \rightarrow C = 7 + 10 = 17$

4.) $S \rightarrow$

5.) $S \rightarrow$

6.) $S \rightarrow$

7.) $S \rightarrow$

8.) $S \rightarrow$

9.) $S \rightarrow$

10.) $S \rightarrow$

11.) $S \rightarrow$

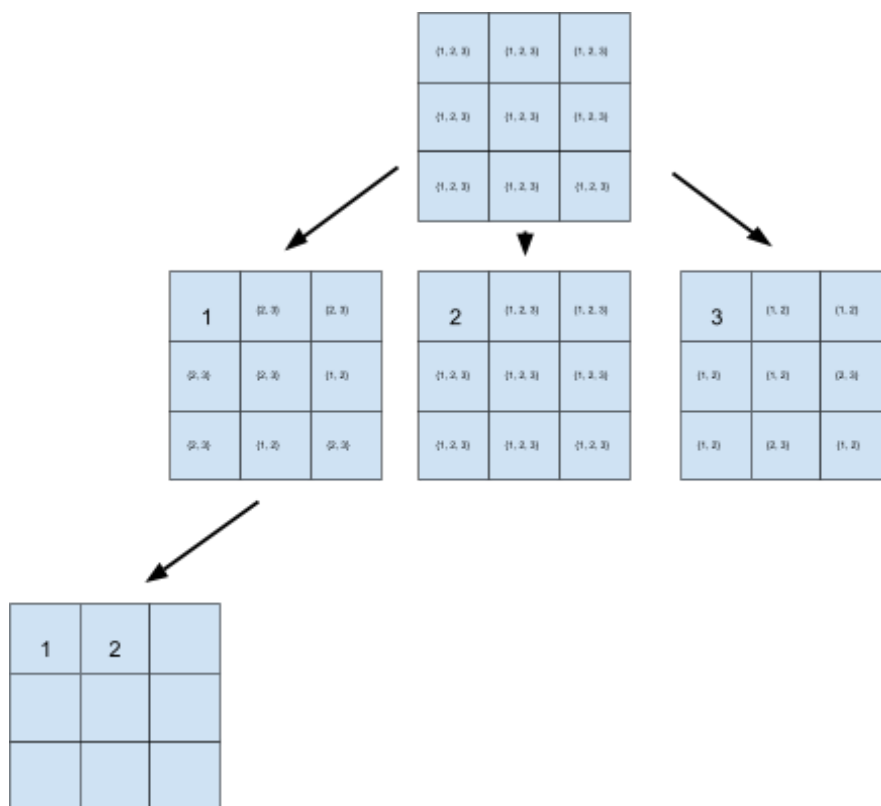
10.) Why would we typically prefer to use DFS rather than BFS for constraint satisfaction problems and backtracking?

11.) Explain in your own words how the backtracking algorithm for CSPs works and is implemented.

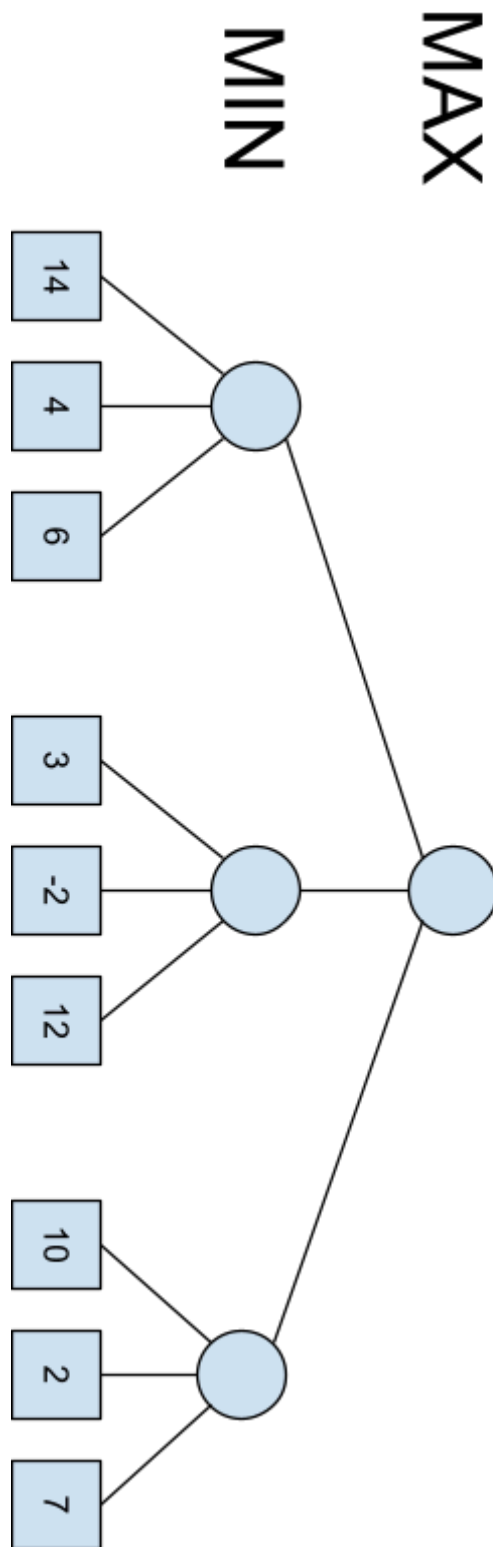
12.) Given the mini-doku problem below, indicate what the AC-3 algorithm would do

1	1, 2, 3 4, 5, 6 7, 8, 9	1, 2, 3 4, 5, 6 7, 8, 9
1, 2, 3 4, 5, 6 7, 8, 9	1, 2, 3 4, 5, 6 7, 8, 9	3
1, 2, 3 4, 5, 6 7, 8, 9	4	1, 2, 3 4, 5, 6 7, 8, 9

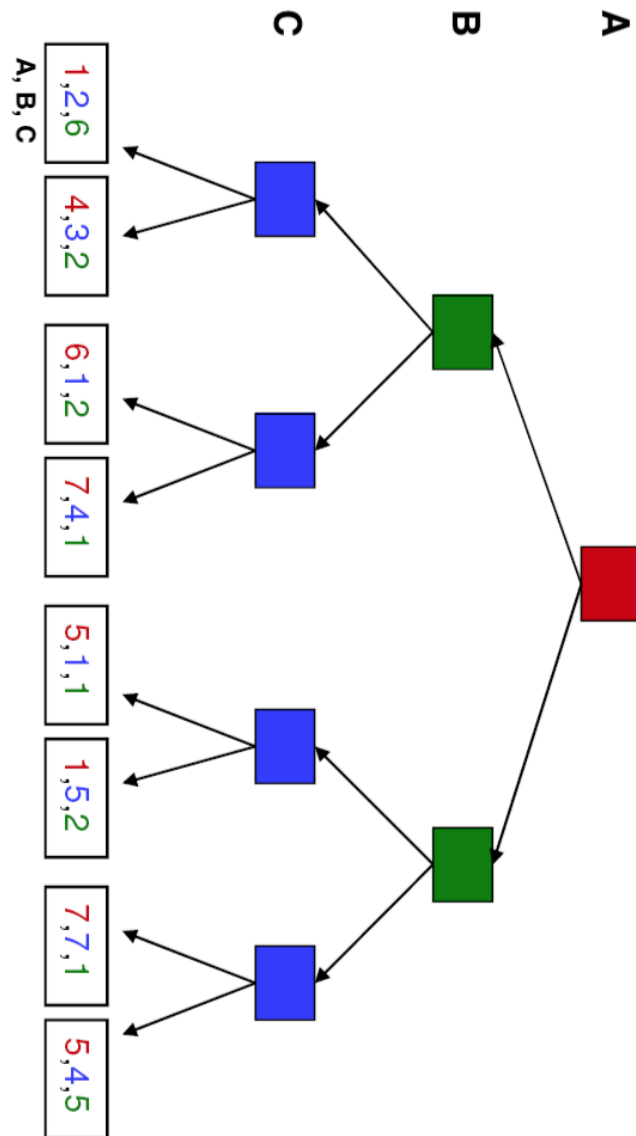
13.) Constraint propagation is when, after making a choice of a value to assign to a variable in a CSP, the newly discovered constraints are propagated forward to future states and their domains. Given below is a problem in which each row, each column, and the upper left to lower right diagonal all need to sum to 6. Each square can have a single value in it from the domain of {1, 2, 3}. Shown below are the first four states in the DFS search with constraint propagation applied. Fill in the missing state.



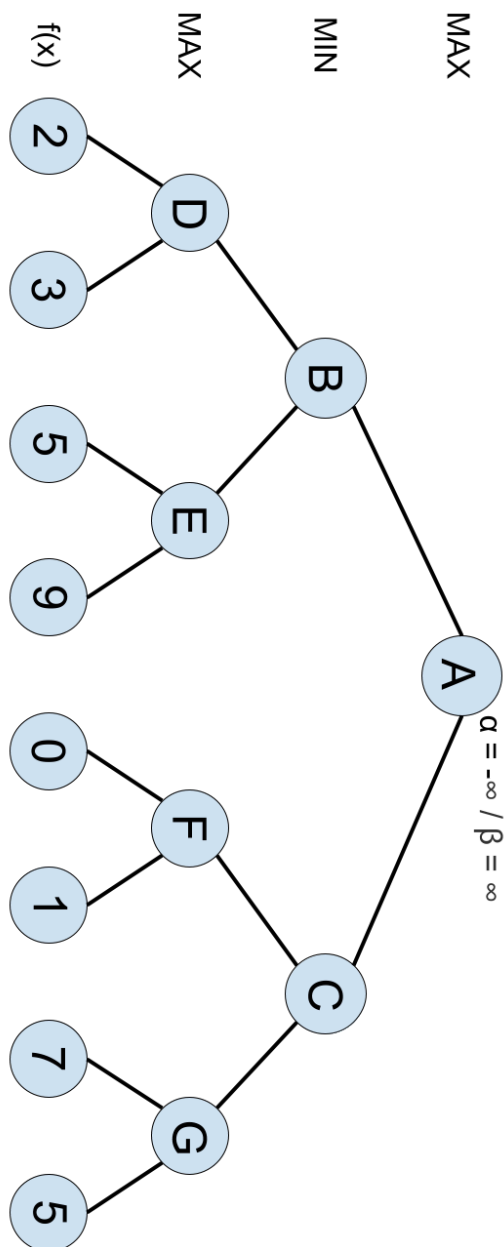
14.) Use Alpha/Beta Pruning on the tree below, showing the Alpha and Beta at each node and indicate which branches don't need to be explored. Make sure to circle the final score.



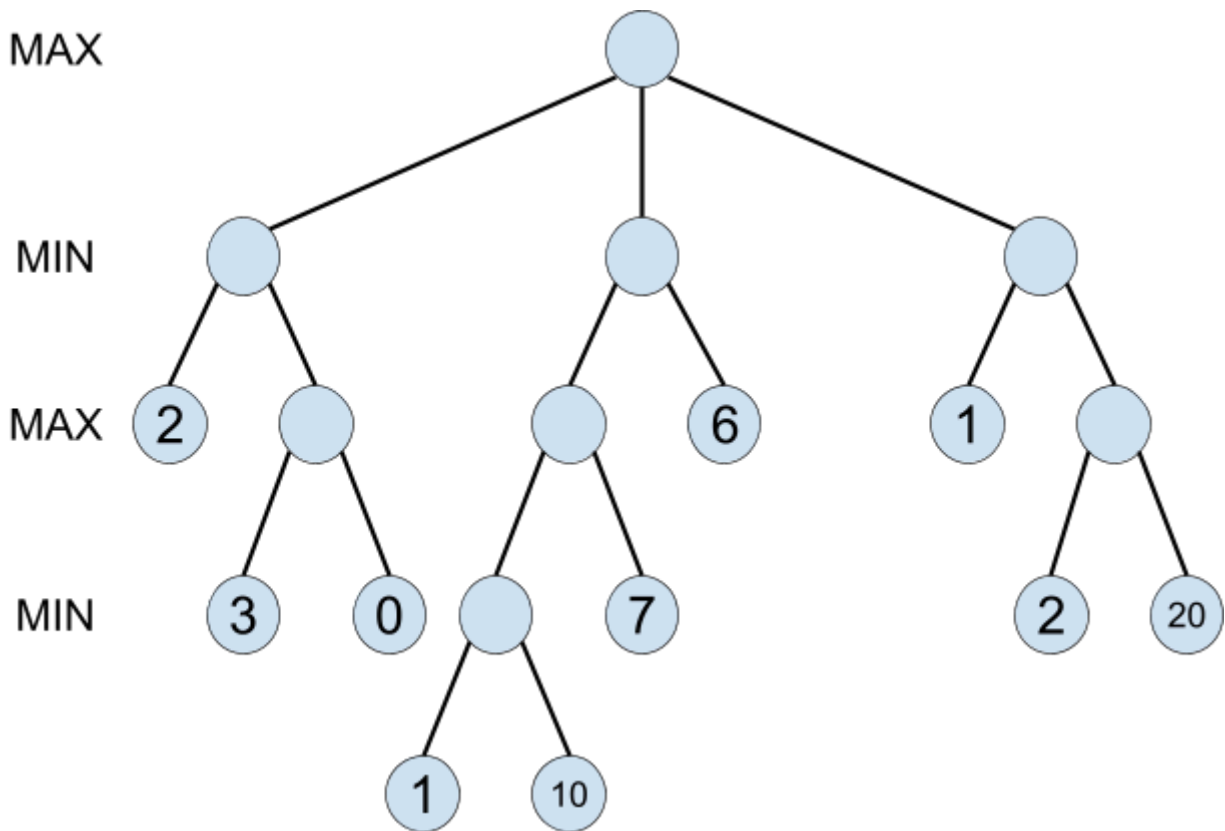
15.) Given the following tree, perform the minimax algorithm for a three-player game. Assume the tuples are ordered in a top-down oriented fashion relative to the players. Show the expected tuple at each empty node of the tree



16.) Use Alpha/Beta Pruning on the tree below, showing the Alpha and Beta at each node and show which branches don't need to be explored.



17.) Given the Minimax tree below, indicate where pruning would occur if you use alpha/beta pruning. **Note:** No need to actually show alpha and beta, but fill in every node with the right value



18.) How do we decide where to split a decision tree?

19.) What are emissions in the context of a Hidden Markov Model? What kind of data would we need to “learn” these?

20.) A weather model is represented as a Markov chain with states: Sunny (S), Rainy (R). Transition probabilities: $P(S \rightarrow S) = 0.8$, $P(S \rightarrow R) = 0.2$, $P(R \rightarrow S) = 0.5$, $P(R \rightarrow R) = 0.5$. If it is Sunny today, what is the probability it will be Sunny two days from now? Also, Explain how the Markov assumption simplifies modeling weather patterns.

21.) Explain the difference between a first-order Markov model and a higher-order Markov model. Give an example of a scenario where a higher-order model would be more appropriate.

22.) Why is the Naive Bayes classifier called “Naive”? What is the big assumption we make while using it?

23.) What is the purpose of the Gaussian distribution in a Naive Bayes classifier?

24.) Given the following dataset layout, answer the two questions.



	S	X		Y
		X_1	X_2	Y
Train	S_1	3.9	4.2	0
	S_2	2.7	8.5	1
	S_3	4.1	4.0	0
	...			
Test	S_{N-1}	3.1	7.9	1
	S_N	4.0	3.8	0

a.) How do we derive the mu and sigma for a gaussian PDF during the training or fitting step of a Naive Bayes classifier?

b.) What is the role of the gaussian PDF during the prediction step of using a Naive Bayes classifier and what assumption do we make by using a Gaussian distribution?

NOTE: I don't expect any math here, just use words and pictures if you like.

25.) What is polynomial feature expansion?

26.) What are the two regularization techniques we learned and how do they affect the weights of a linear regression?

27.) What would the equation look like for a multiple linear regression with three features?

28.) In ordinary least squares regression, what exactly are we minimizing when we choose the “best fit” line, and why do we square the residuals instead of just summing them?

29.) Suppose we have data points $(x,y) = \{(1,2),(2,3),(3,5)\}$. Write the linear regression model and set up (but do not fully solve) the OLS minimization problem for w_0 and w_1 .

30.) What is a loss function?

31.) What is the purpose of the activation function in the perceptron?

32.) What is the purpose of the weight update function (beyond just updating the weights) in the perceptron?

33.) Given below is a Perceptron. Using the Step function between -1 and 1 as the activation function, classify the three points given below. Then update the Perceptron weights for each misclassified sample and record the new weights.

$$\omega = (1.2, 0.7, 0.2, 0.5), \eta = 0.2$$

Samples	X_1	X_2	X_3	Y
S_1	2	3	-1	1
S_2	1	1	1	1
S_3	-2	-4	3	-1

Classify:

S_1 :

S_2 :

S_3 :

Weight Updates:

Final Weights:

$$\omega = (\omega_0: \quad, \omega_1: \quad, \omega_2: \quad, \omega_3: \quad)$$

34.) What are the differences between classification and regression tasks? Give an example of each

35.) What is the purpose of regularization and how are L1 and L2 regularization different?

36.) What is the bias/variance trade-off?

37.) Why is it important to separate our dataset into training and testing data?

38.) In your own words, briefly describe the following terms and explain the difference between them: Artificial Intelligence, Machine learning, Deep learning

39.) What are over and underfitting?

40.) What is a feature vector, why are these useful?

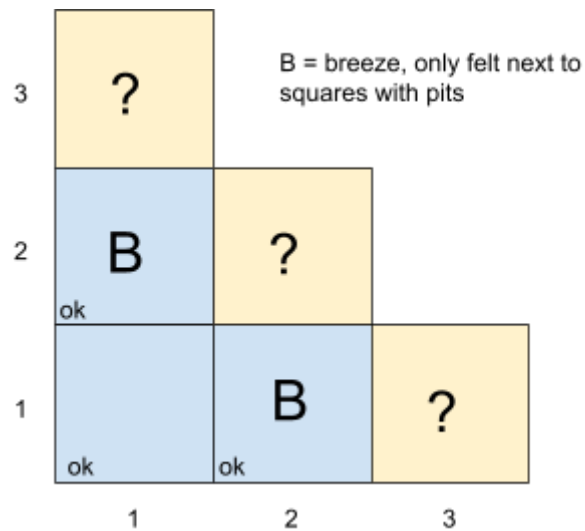
41.) What are MLE and MAP?

42.) How are minimax and alpha/beta pruning related? Do we still have to use minimax if we use alpha/beta pruning?

43.) Write the following statement:

"I must always split my data into training and testing and must not train on the testing data"

44.) Given the wumpus world answer the following questions



a.) What would a logic based agent conclude about the three squares with question marks in them ([1,3], [2,2], [3,1])? Which square would a logical agent move into next? Would it move at all?

b.) Which of the squares do you personally think is the most likely to contain the pit(s)? Explain your reasoning. If you had to pick one of the three to move into, which would you choose?

